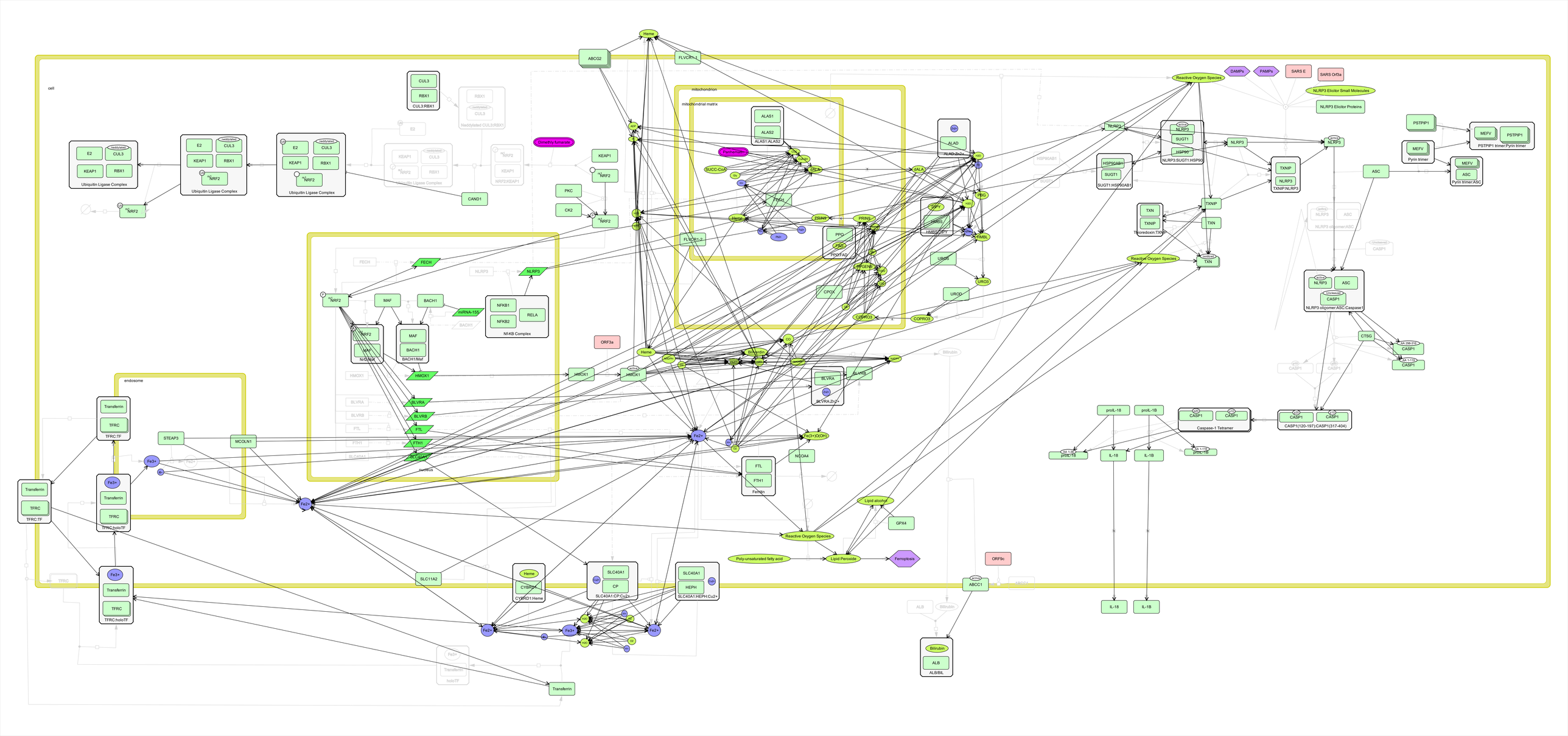

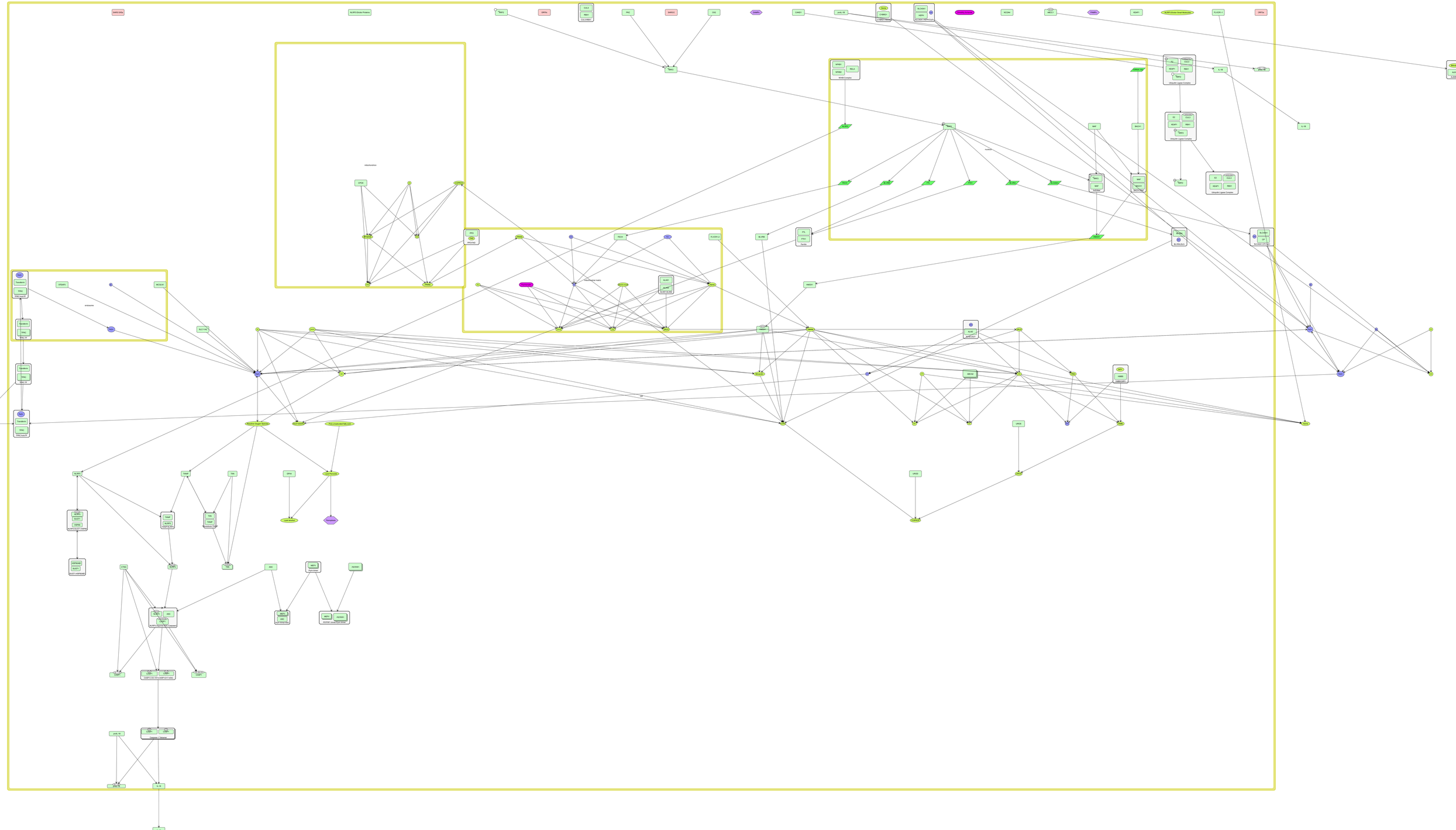

Heme
ABCG2
FLVCR1-1
cell
CUL3
RBX1
CUL3:RBX1
RBX1
neddylated
CUL3
Neddylated CUL3:RBX1
E2
neddylated
CUL3
KEAP1
RBX1
Ubiquitin Ligase Complex
E2
neddylated
CUL3
KEAP1
RBX1
Ub
rs1 NRF2
Ubiquitin Ligase Complex
Ub
E2
neddylated
CUL3
KEAP1
RBX1
rs1 NRF2
Ubiquitin Ligase Complex
Ub
E2
KEAP1
neddylated
CUL3
rs1 NRF2
RBX1
Ubiquitin Ligase Complex
rs1 NRF2
KEAP1
NRF2:KEAP1
Ub
rs1 NRF2
CAND1
Dimethly fumarate
KEAP1
rs1 NRF2
PKC
CK2
P
rs1 NRF2
ADP
Pi
H2O
ATP
mitochondrion
mitochondrial matrix
ALAS1
ALAS2
ALAS1:ALAS2
Panhematin
CO2
CoA-SH
SUCC-CoA
Gly
dALA
H+
FECH
Heme
PRIN9
H+
Pb2+
Fe2+
PRIN9
PPO
FAD
PPO:FAD
?
dALA
Zn2+
ALAD
ALAD:Zn2+
H2O
H+
PBG
DIPY
HMBS
HMBS:DIPY
H2O
NH4+
HMBL
H2O2
O2
PPGEN9
H2O2
CO2
UROS
URO3
UROD
CPOX
O2
COPRO3
COPRO3
FLVCR1-2
FECH
FECH
NLRP3
NLRP3
P
rs1 NRF2
MAF
BACH1
miRNA-155
BACH1
NFKB1
NFKB2
RELA
Nf-KB Complex
P
rs1 NRF2
MAF
Nrf2/Maf
MAF
BACH1
BACH1/Maf
HMOX1
HMOX1
BLVRA
BLVRA
BLVRB
BLVRB
FTL
FTL
FTH1
FTH1
SLC40A1
SLC40A1
nucleus
ORF3a
HMOX1
active
HMOX1
Heme
NADPH
O2
NADP+
Biliverdin
H2O
NADPH
CO
NADP+
Bilirubin
BLVRA
Zn2+
BLVRA:Zn2+
BLVRB
Fe2+
H+
O2
Fe(3+)O(OH)
NCOA4
FTL
FTH1
Ferritin
Lipid alcohol
GPX4
Reactive Oxygen Species
Poly-unsaturated fatty acid
Lipid Peroxide
Ferroptosis
endosome
Transferrin
TFRC
TFRC:TF
STEAP3
MCOLN1
Fe3+
Fe2+
e-
Fe2+
Transferrin
TFRC
TFRC:TF
Fe3+
Transferrin
TFRC
TFRC:holoTF
TFRC
Fe3+
Transferrin
TFRC
TFRC:holoTF
SLC11A2
Heme
CYBRD1
CYBRD1:Heme
SLC40A1
CP
Cu2+
SLC40A1:CP:Cu2+
SLC40A1
HEPH
Cu2+
SLC40A1:HEPH:Cu2+
Fe2+
e-
Fe3+
H2O
H+
O2
H2O
O2
H+
Fe2+
Fe3+
Transferrin
holoTF
Transferrin
ORF9c
active
ABCC1
ABCC1
ALB
Bilirubin
Bilirubin
ALB
ALB/BIL
Reactive Oxygen Species
DAMPs
PAMPs
SARS E
SARS Orf3a
NLRP3 Elicitor Small Molecules
NLRP3 Elicitor Proteins
NLRP3
active
NLRP3
SUGT1
HSP90
NLRP3:SUGT1:HSP90
HSP90AB1
SUGT1
HSP90AB1
SUGT1
SUGT1:HSP90AB1
NLRP3
active
NLRP3
TXNIP
NLRP3
TXNIP:NLRP3
TXNIP
TXN
TXNIP
Thioredoxin:TXNIP
TXN
Reactive Oxygen Species
oxidized
TXN
PSTPIP1
MEFV
PSTPIP1
PSTPIP1 trimer:Pyrin trimer
MEFV
Pyrin trimer
ASC
MEFV
ASC
Pyrin trimer:ASC
active
NLRP3
ASC
NLRP3 oligomer:ASC
Uncleaved
CASP1
active
NLRP3
ASC
Uncleaved
CASP1
NLRP3 oligomer:ASC:Caspase1
CTSG
AA 298-316
CASP1
AA 1-119
CASP1
p10
CASP1
p20
CASP1
proIL-18
proIL-1B
p20
CASP1
p10
CASP1
Caspase-1 Tetramer
p20
CASP1
p10
CASP1
CASP1(120-197):CASP1(317-404)
AA 1-36
proIL-18
IL-18
IL-1B
AA 1-116
proIL-1B
IL-18
IL-1B

SARS Orf3a
NLRP3 Elicitor Proteins
NRF2
ORF9c
CUL3
RBX1
CUL3:RBX1
PKC
SARS E
CK2
DAMPs
CAND1
proIL-1B
Heme
CYBRD1
CYBRD1:Heme
SLC40A1
HEPH
Cu2+
SLC40A1:HEPH:Cu2+
Dimethly fumarate
NCOA4
active
ABCC1
PAMPs
KEAP1
NLRP3 Elicitor Small Molecules
FLVCR1-1
ORF3a
mitochondrion
CPOX
O2
COPRO3
PPGEN9
CO2
H2O
PRIN9
PPO
FAD
PPO:FAD
NRF2
NFKB1
NFKB2
RELA
Nf-KB Complex
NLRP3
NRF2
nucleus
FECH
BLVRB
FTL
FTH1
BLVRA
SLC40A1
MAF
BACH1
miRNA-155
NRF2
MAF
Nrf2/Maf
MAF
BACH1
BACH1/Maf
HMOX1
Ub
E2
neddylated
CUL3
KEAP1
RBX1
NRF2
Ubiquitin Ligase Complex
E2
CUL3
KEAP1
RBX1
NRF2
Ubiquitin Ligase Complex
NRF2
E2
CUL3
KEAP1
RBX1
Ubiquitin Ligase Complex
IL-1B
AA 1-116
proIL-1B
IL-1B
Bilirubin
ALB
ALB/BIL
PRIN9
Fe2+
FECH
Pb2+
FLVCR1-2
BLVRB
FTL
FTH1
Ferritin
Gly
Panhematin
H+
mitochondrial matrix
SUCC-CoA
ALAS1
ALAS2
ALAS1:ALAS2
Heme
CoA-SH
CO2
dALA
HMOX1
BLVRA
Zn2+
BLVRA:Zn2+
SLC40A1
CP
Cu2+
SLC40A1:CP:Cu2+
Fe3+
Transferrin
TFRC
TFRC:holoTF
STEAP3
e-
MCOLN1
endosome
Transferrin
TFRC
TFRC:TF
Fe3+
SLC11A2
O2
NADPH
active
HMOX1
Heme
Zn2+
ALAD
ALAD:Zn2+
dALA
e-
Fe2+
H+
O2
Fe2+
CO
Biliverdin
H+
ATP
ABCG2
PBG
DIPY
HMBS
HMBS:DIPY
Fe3+
H2O
Transferrin
TFRC
TFRC:TF
cell
Transferrin
Fe3+
Transferrin
TFRC
TFRC:holoTF
Reactive Oxygen Species
Fe(3+)O(OH)
Poly-unsaturated fatty acid
NADP+
ADP
Pi
NH4+
HMBL
Heme
UROS
NLRP3
TXNIP
TXN
GPX4
Lipid Peroxide
UROD
URO3
active
NLRP3
SUGT1
HSP90
NLRP3:SUGT1:HSP90
TXNIP
NLRP3
TXNIP:NLRP3
TXN
TXNIP
Thioredoxin:TXNIP
Lipid alcohol
Ferroptosis
COPRO3
HSP90AB1
SUGT1
SUGT1:HSP90AB1
CTSG
active
NLRP3
oxidized
TXN
ASC
MEFV
Pyrin trimer
PSTPIP1
active
NLRP3
ASC
uncleaved
CASP1
NLRP3 oligomer:ASC:Caspase1
MEFV
ASC
Pyrin trimer:ASC
MEFV
PSTPIP1
PSTPIP1 trimer:Pyrin trimer
AA 1-119
CASP1
p20
CASP1
p10
CASP1
CASP1(120-197):CASP1(317-404)
AA 298-316
CASP1
proIL-18
p20
CASP1
p10
CASP1
Caspase-1 Tetramer
AA 1-36
proIL-18
IL-18
IL-18